

PREPARING FOR NEGATIVE RATES



**AND OTHER POSSIBILITIES IN THIS
UNCONVENTIONAL ENVIRONMENT**

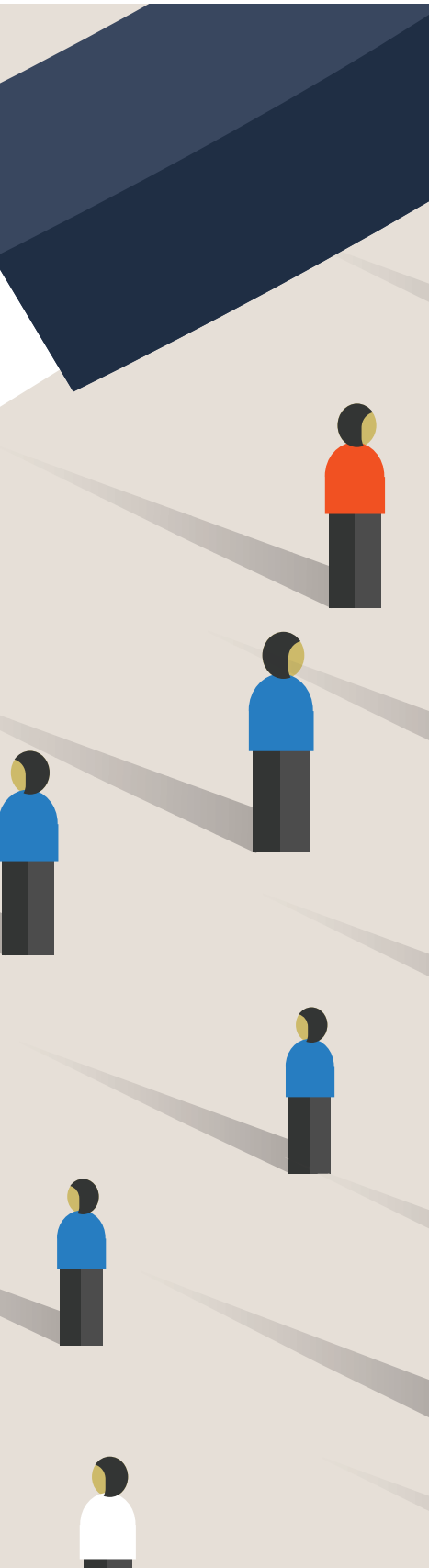


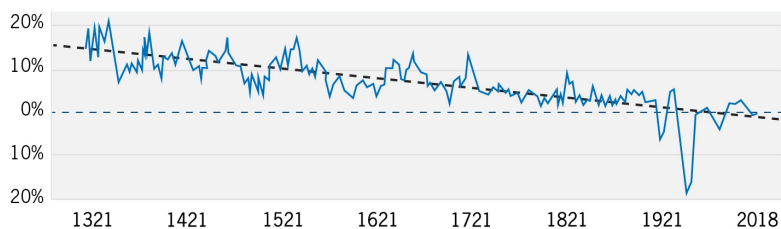
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BY JOSEPH A. IRACI

NEGATIVE INTEREST RATES are one of the more unconventional monetary tools at the disposal of central bankers. As the table below shows, real interest rates—those that account for inflation—have been declining for several centuries, and negative real interest rates have been experienced well before the 2008 recession.

Real rates have been in decline since the 14th century Black Death

Real interest rate for personal loans, 10-year moving average.



SOURCE: *Eight centuries of global real interest rates* from the Bank of England

In theory, negative interest rates should incentivize spending and stimulate the economy by encouraging consumers and financial firms to borrow and lend more. Negative rates should be a disincentive to hold cash as deposits by making it costly to do so. However, the use of negative rates has come with mixed results. Central bankers remain divided on their usefulness to stimulate economic growth.

Negative interest rates garnered a lot of attention in 2014 when the European Central Bank (ECB) set the Deposit Rate Facility below zero. It was the first time negative rates were used to stimulate an economy. Prior to the ECB, the Bank of Japan (BOJ) adopted a zero-interest rate policy in 1999 and its key rate has been negative since 2016. As of October 2020, five central banks were employing negative interest rates: ECB, BOJ, Switzerland, Denmark, and Sweden. In addition to having mixed stimulative results, negative rates present challenges regarding pricing and managing the risk of derivative instruments and bonds. Most models and systems assume a non-negative rate.

Monetary Policy

Central bankers differ on using negative rates as a monetary policy mechanism. Jerome Powell, Chairman of the Federal Reserve, recently said,

"I continue to think, and my colleagues on the Federal Open Market Committee continue to think that negative interest rates is probably not an appropriate or useful policy for us here in the United States....The evidence on whether it helps is quite mixed."

Still, while negative rates remain an unconventional tool, unconventional has become the new normal as central banks take more aggressive action across the globe.

The global financial crisis of 2007-09 had very different underlying causes than the crisis and volatility prompted by COVID-19. In both cases, central banks acted swiftly to maintain financial stability and ease monetary policy, but their recent actions have surpassed their responses to the previous crisis.

The spreading coronavirus has disrupted business activity and created uncertainty, which reduced asset prices

around the world and prompted investors to sell risky assets and buy safer assets, such as U.S. Treasury securities. This was particularly true from March 9 to March 26, 2020, when financial market conditions resembled those of late 2008. These sales of risky assets pushed those prices down further, reducing the net worth of potential borrowers, which made lenders reluctant to lend to these less creditworthy borrowers. By making short-term credit available at a modest cost, absorbing credit risk, and reducing the cost of long-term borrowing, central banks sought to stop the cycle of fire-sale declines in asset prices that were further reducing lending and business activity.

Policy rates in 2020 were not far above zero by historical standards, whereas the rates prior to the 2007-09 crisis started out well above the levels in early 2020. With short-term policy rates constrained from below, central banks again turned to purchasing long-term bonds to reduce their yields and to lending programs to facilitate credit in specific sectors. Central bank balance sheets have consequently expanded at record pace.

Since the global financial crisis, asset purchase programs have become a staple in the arsenal of central banks, especially with short rates near their zero-lower bound. Unlike other central banks, the Fed has announced it will conduct an open-ended purchase program, buying Treasuries and agency mortgage-backed securities. The Fed is buying assets unusually quickly compared with past purchase episodes. Treasury purchases peaked at \$75 billion in late March before settling into the recent pace of \$15 billion per day, which is much quicker than the peak pace of \$120 billion per month during the prior financial crisis. These purchases have caused spikes in the Fed's balance sheet.

In addition to restarting some facilities from the 2007-09 crisis, the Fed now intends to support lending to municipalities, corporations, and small and medium-sized enterprises (SMEs).

These Fed lending facilities may jointly lend more than \$2.3 trillion.

The Fed in addition to the three other main central banks (BOJ, ECB, Bank of England) also relaxed bank capital requirements to foster lending. The ECB and Bank of England (BOE) have requested that banks halt or limit dividends, buybacks, and bonuses. The Fed has expanded standing U.S. dollar liquidity swaps and created a repo facility to facilitate dollar-denominated lending by international monetary authorities.

The scope and speed of the international central bank policy response to COVID-19 has met or often exceeded that of 2007-09. Central banks have cut rates, purchased assets, created lending programs, and modified financial regulations to support financial markets and minimize long-run economic damage.

This central bank intervention to either drive negative interest rates or to ensure rates remain low could have several consequences:

- Lenders expect an interest-bearing return based on the risk of the borrower; negative interest rates distort the risk/return relationship.
- Negative rates could cause bank runs due to negative-yielding savings deposits, and behavioral changes could mitigate the stimulative impact of negative or low rates.
- Increased central bank intervention combined with potential behavioral changes could impact global growth and interest rates as the global economy emerges from the coronavirus. Financial institutions must be prepared for any interest rate scenario.

Interest Rates

A number of models are used to forecast interest rates. Many are based on quantitative data but have the ability for a qualitative adjustment. However, as we continue to experience increased central bank intervention to the point where unconventional is now conventional monetary policy, combined with

the potential for behavioral changes by global consumers as economies come out of COVID-19, it seems reasonable to begin deploying scenario analysis and considering operational risk when forecasting interest rates.

- In "Longer-Run Economic Consequences of Pandemics," (<https://www.frbsf.org/economic-research/files/wp2020-09.pdf>) Oscar Jorda, Sanjay Singh, and Alan Taylor determined that pandemics depress economic growth for several decades following the pandemic.
- In "Scarring Body and Mind: The Long-Term Belief – Scarring Effects of COVID-19," (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3588480) Julian Kozlowski, Laura Veldkamp, and Venky Venkateswaran took the position that the coronavirus could leave economic scars.

Operational risk is concerned with people, process, systems, and external events. External events can range from geopolitical risk, climate change, monetary policies, fiscal policies, and behavioral changes by consumers that could mitigate the effects of monetary and fiscal stimulus. If external events have the potential to impact economic growth and interest rates, they may need to be taken into consideration when modeling or forecasting interest rates.

Bayesian networks can be used to model operational risk. To do this, the external environment needs to be evaluated to determine potential threats or opportunities. The internal environment also needs to be evaluated to determine the strength of management, internal controls, and the effectiveness of analysis and reporting.

After the external and internal environments are assessed the net potential impact can be analyzed. A Bayesian network is a multi-variate statistical model that can incorporate quantitative and qualitative inputs.

Models, including stress tests, continue to be important tools. However, a scenario analysis that includes opera-



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tional risk can be used to assess and evaluate potential future events that are extreme but plausible. These events are not necessarily derived from historical events, although history can be a starting point when building a scenario. The purpose of a scenario analysis is to:

- Prepare for unexpected events.
- Identify key risk exposures, especially emerging risks.
- Explore potential extreme events.
- Mitigate risk of solvency.

The recent pandemic supports the development of scenario analysis to supplement models because risk manifesting in one area tends to spread to other risk classifications. To review the timeline of COVID-19:

- January 9: The World Health Organization (WHO) announces coronavirus pneumonia in Wuhan, China.
- January 31: WHO issues a Global Health Emergency.
- March 11: WHO declares COVID-19 a pandemic.

All of the above is considered an operational risk event, but this event drove economic growth concerns, resulting in a severe global market downturn starting in February 2020. Additionally, banks began taking provisions to bolster loan loss reserves to prepare for possible defaults and bankruptcies. At the same time, we witnessed unprecedented monetary and fiscal

stimulus across the globe to combat the economic impact of the pandemic.

An operational risk event led to economic growth concerns, which led to heightened market risk as assets priced lower, which led to credit risk concerns, and monetary and fiscal stimulus was used to offset the negative economic impact of the pandemic, restore confidence, and inject liquidity. But now we are seeing research beginning to be published on behavioral changes that have the potential to impact growth and policies going forward, and the pandemic is just the most recent operational risk event that has caused heightened market, credit, and liquidity risks.

Scenario analysis can be used to prepare for extreme events, which are happening with increased frequency, because it can handle multiple risk factors. Models, meanwhile, including stress tests, are better suited when there are fewer risk factors to consider.

Conclusion

Central banks have become increasingly creative to combat disinflation and provide economic stimulus. Governments have also provided unprecedented fiscal stimulus. One of the consequences of monetary and fiscal policies, including negative in-

terest rates, is that asset prices could be distorted as intervention impacts market pricing. While intervention is needed to restore confidence in the global economy, it has the potential to cause anomalies in asset prices and interest rates. External events, whether driven by geopolitical risk, behavioral economics, monetary / fiscal policies, or something unforeseen, need to be included when analyzing extreme but plausible events to ensure firms are positioned to operate in an increasingly volatile operating environment. [®]



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